

Qatar-10b

R-0079

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_241022 · created 2026-07-15T22:13:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460605.6604 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.021
Transit depth (Rp/Rs)^2	1.81 +/- 0.56 [%]
Orbital Inclination (inc)	88.37 +/- 2.5
Airmass coefficient 1 (a1)	1.287 +/- 0.023
Airmass coefficient 2 (a2)	-0.151 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.78 %
Best Comparison Star	None
Optimal Aperture	2.73
Optimal Annulus	9.07
Transit Duration (day)	0.1194 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.021 vs 0.1265 (deviation 0.36σ)
Duration fit vs published	0.1194 d vs 0.1154 d (deviation 0.78σ)
Mid-time O–C	0.2 min
Depth SNR	6.4
Residual scatter	1.78 %

Light curve

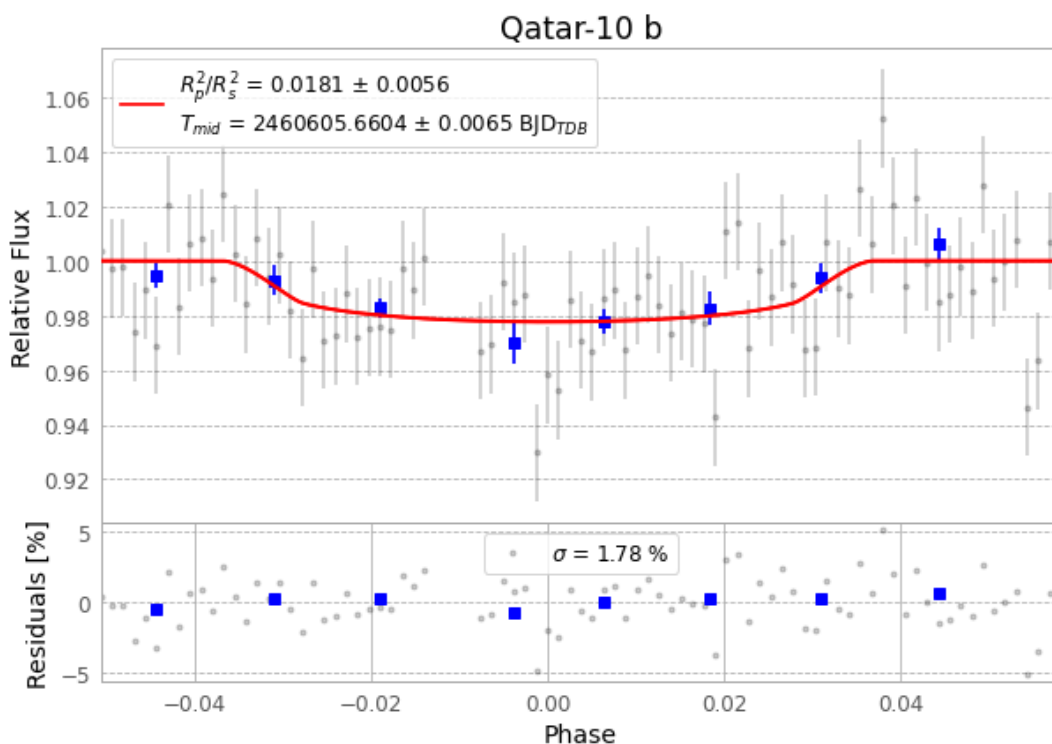


Figure 1 — FinalLightCurve_Qatar-10 b_2024-10-21.png (SHA-256 d9b4452962817703...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a981c5a80002477...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

87 FITS frames (57 MB), Qatar-10241022014815.FITS → Qatar-10241022060615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-241022.log (7b32f7fafa05...), FinalLightCurve_Qatar-10 b_2024-10-21.png (d9b445296281...), FinalLightCurve_Qatar-10 b_2024-10-21.csv (8b16db5e12ec...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:13Z.